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CSCE 581

NCAA Women's Basketball Game Outcome Prediction Models

Overview: An increase in parity across women's basketball teams in recent years has both improved the "product" and entertainment value of women's basketball. This parity across teams motivated me to explore model performance in predicting game outcome, as there are now many strong teams and the odds are often very close for determining outcome. This project explores predictive probability of game outcomes for division 1 NCAA women's basketball games using team identity and performance metrics. Both random forest and logistic regression were implemented and examined for model performance as well as feature importance for decision making.

Trust Issues: There are a few trust issues both in regards to the overall context of this issue, as well as the specific data and models implemented. Because of a historic lack of parity in women's college basketball, historically dominant teams are at risk of being systemically over valued with lack of considerations for current team variables. Additionally, the feature set is missing some context that may be helpful in making meaningful predictions, context that gives analysts an edge over these models if there are player injuries, significant strength of schedule disparities, and other context that is unaccounted for.

Data Engineering:

Game stats, including field goal percentage, 3 point percentage, points, rebounds, assists, and more are correlated directly to win-loss. This caused data leakage in early models, and therefore team statistics before the game are a more relevant metrics for prediction. Data was engineered to calculate *games played before date of game* and *win percentage before date of game* for both team and opponent. Additionally, due to the data format (exported from https://www.sports-reference.com/) each game was duplicated, with one version for each team as "team" and the other as "opponent". After removing duplicates, the winning team was always referenced as "team" and the losing team was referenced as "opponent" which caused random forest and logistic regression results to be inflated. I fixed this issue by reformatting the data to reference the teams alphabetically, which was successful in balancing the data.

Data Summary:

Original Dataset:

							Team														Opponent												
Rk	Team	Date	PTS		Opp	Result	MP	FG	FGA	FG%	2P	2PA	2P%	3P	3PA	3P%	FT	FTA	FT%	PTS	FG	FGA	FG%	2P	2PA	2P%	3P	3PA	3P%	FT	FTA	FT%	PTS
1	Vermont	2025-11-04	137		Dean	W 137-31	200	56	82	.683	43	57	.754	13	25	.520	12	16	.750	137	11	47	.234	7	30	.233	4	17	.235	5	10	.500	31
2	Ga. Southern	2025-12-12	136		Coastal Georgia	W 136-74	200	50	73	.685	35	45	.778	15	28	.536	21	29	.724	136	26	77	.338	13	34	.382	13	43	.302	9	15	.600	74
3	Charleston	2025-11-03	135		Covenant	W 135-41	200	51	93	.548	41	64	.641	10	29	.345	23	27	.852	135	11	46	.239	8	20	.400	3	26	.115	16	20	.800	41
4	Oklahoma St.	2025-12-06	133		Mississippi Valley State	W 133-46	200	45	67	.672	30	36	.833	15	31	.484	28	32	.875	133	17	58	.293	14	47	.298	3	11	.273	9	12	.750	46
5	Saint Louis	2025-12-28	132		Stl Pharmacy	W 132-53	200	49	85	.576	32	47	.681	17	38	.447	17	22	.773	132	19	71	.268	15	49	.306	4	22	.182	11	13	.846	53
6	Troy	2025-12-10	132		Oakwood	W 132-62	200	49	85	.576	29	50	.580	20	35	.571	14	22	.636	132	22	67	.328	14	39	.359	8	28	.286	10	17	.588	62
7	Ohio St.	2025-11-30	130		Niagara	W 130-32	200	52	85	.612	42	61	.689	10	24	.417	16	24	.667	130	10	63	.159	8	46	.174	2	17	.118	10	12	.833	32
8	Longwood	2025-11-14	128		Bluefield Univ.	W 128-31	200	46	94	.489	36	69	.522	10	25	.400	26	37	.703	128	14	48	.292	12	35	.343	2	13	.154	1	2	.500	31
9	Oklahoma	2025-12-22	126		North Carolina Central	W 126-54	200	48	88	.545	34	58	.586	14	30	.467	16	20	.800	126	15	64	.234	10	46	.217	5	18	.278	19	29	.655	54
10	LSU	2025-12-07	126	@	New Orleans	W 126-62	200	50	85	.588	46	71	.648	4	14	.286	22	30	.733	126	19	62	.306	11	36	.306	8	26	.308	16	21	.762	62
11	Wofford	2025-11-03	126		Converse	W 126-65	200	48	82	.585	39	61	.639	9	21	.429	21	28	.750	126	17	57	.298	10	23	.435	7	34	.206	24	32	.750	65
12	Santa Clara	2025-12-06	125		Simpson (CA)	W 125-43	200	44	93	.473	31	48	.646	13	45	.289	24	29	.828	125	14	54	.259	11	43	.256	3	11	.273	12	18	.667	43
13	Michigan St.	2025-11-04	125		Mercyhurst	W 125-39	200	49	79	.620	36	51	.706	13	28	.464	14	20	.700	125	12	35	.343	6	15	.400	6	20	.300	9	10	.900	39
14	Sacramento St.	2025-11-03	124		Stanton	W 124-39	200	52	86	.605	45	63	.714	7	23	.304	13	22	.591	124	14	41	.341	5	17	.294	9	24	.375	2	5	.400	39
15	Texas	2025-11-03	123		Incarnate Word	W 123-51	200	51	78	.654	44	61	.721	7	17	.412	14	17	.824	123	13	47	.277	10	30	.333	3	17	.176	22	32	.688	51
16	FIU	2025-12-30	122		Florida National	W 122-30	200	48	91	.527	38	58	.655	10	33	.303	16	32	.500	122	9	31	.290	6	24	.250	3	7	.429	9	12	.750	30
17	TCU	2025-11-12	122		Tennessee State	W 122-39	200	40	68	.588	23	41	.561	17	27	.630	25	34	.735	122	16	57	.281	13	41	.317	3	16	.188	4	9	.444	39
18	Wright St.	2025-11-12	122		Wilbeforce	W 122-47	200	43	77	.558	25	39	.641	18	38	.474	18	23	.783	122	17	60	.283	16	44	.364	1	16	.063	12	21	.571	47
19	South Carolina	2025-11-23	121		Queens (NC)	W 121-49	200	43	65	.662	35	45	.778	8	20	.400	27	34	.794	121	16	70	.229	7	42	.167	9	28	.321	8	10	.800	49
20	UMKC	2025-11-20	121		Hesston College	W 121-43	200	45	74	.608	34	58	.586	11	16	.688	20	31	.645	121	14	52	.269	7	29	.241	7	23	.304	8	14	.571	43
21	Louisiana Tech	2025-11-11	121		La. Christian	W 121-34	200	50	98	.510	38	66	.576	12	32	.375	9	13	.692	121	11	60	.183	7	33	.212	4	27	.148	8	8	1.000	34
22	Longwood	2025-11-05	121		Randolph	W 121-35	200	47	87	.540	34	59	.576	13	28	.464	14	17	.824	121	14	57	.246	11	42	.262	3	15	.200	4	12	.333	35
23	Texas	2025-12-28	120		Southeastern Louisiana	W 120-38	200	48	80	.600	40	58	.690	8	22	.364	16	25	.640	120	15	45	.333	14	41	.341	1	4	.250	7	15	.467	38
24	Fla. Gulf Coast	2025-12-22	120		Ft. Lauderdale	W 120-32	200	49	73	.671	38	48	.792	11	25	.440	11	15	.733	120	12	53	.226	8	34	.235	4	19	.211	4	14	.286	32
25	Michigan	2025-11-18	120		Binghamton	W 120-50	200	45	68	.662	34	48	.708	11	20	.550	19	29	.655	120	17	50	.340	12	31	.387	5	19	.263	11	16	.688	50
26	Texas Southern	2025-12-29	119		College of Biblical Studies	W 119-58	200	47	90	.522	34	60	.567	13	30	.433	12	15	.800	119	19	63	.302	12	33	.364	7	30	.233	13	18	.722	58
27	UNCG	2025-12-13	119		Bob Jones	W 119-35	200	45	81	.556	32	54	.593	13	27	.481	16	20	.800	119	14	47	.298	10	30	.333	4	17	.235	3	4	.750	35
28	Ball St.	2025-12-03	119		Oakland City	W 119-34	200	49	96	.510	39	60	.650	10	36	.278	11	14	.786	119	11	54	.204	8	35	.229	3	19	.158	9	14	.643	34
29	Iowa	2025-11-09	119		Evansville	W 119-43	200	42	65	.646	28	39	.718	14	26	.538	21	28	.750	119	16	61	.262	10	37	.270	6	24	.250	5	7	.714	43
30	West Virginia	2026-03-01	118		Cincinnati	W 118-60	200	42	72	.583	27	42	.643	15	30	.500	19	30	.633	118	22	53	.415	20	42	.476	2	11	.182	14	24	.583	60

Transformed Dataset Example:

Team	**Opponent**	**Date**	**Team Games Before Date**	**Opponent Games Before Date**	**Team Win Percentage**	**Opponent Win Percentage**
Baylor	UCF	1/21/2026	20	18	0.85	0.55
BYU	Texas Tech	1/21/2026	18	20	0.77	0.95
Butler	DePaul	1/21/2026	19	10	0.42	0.20
Providence	Seton Hall	1/20/2026	20	18	0.50	0.72

Model Details: The initial data set, after removing duplicate games, contains 6024 unique game items. An 80/20 train test split was used, which produced a test set of 1206 games. Input features for both models include team, team games before date, team win percentage before date, opponent, opponent games before date, and opponent win percentage before date. Conversion from non numeric features, team and opponent, was done using one hot encoding.

For logistic regression, initial hyperparameters C=1e5 and max iterations=1000 led to convergence issues. Reducing C and increasing max iterations successfully allowed the model to converge and produce meaningful results.

Logistic Regression F1 Score: 0.740

Logistic Regression AUC ROC: 0.819

Random forest hyperparameters were initially set to max depth=2 and random state=0 and did not need to be altered.

Random Forest F1 score: 0.639

Random Forest AUC ROC: 0.778

Logistic regression likely performed better due to a simple feature set where a linear model is favored. Additionally, the maximum depth hyperparameter for random forest is quite low, so an increase may improve model performance but could also cause over-fitting issues.

Evaluation Metrics: Confusion matrix, F1 score, AUC ROC, and comparison to LLM performance were all used to evaluate random forest and logistic regression models. F1 score and AUC ROC both indicate that logistic regression exhibited better performance than random forest. The LLM (Chat GPT) outperformed both models with an F1 score marginally higher than the F1 score for logistic regression, but was less balanced with significantly more false positive values than false negative values.

Explanation Methods: Both LIME and SHAP were used to explain results. While these explanation methods are useful in understanding how the models determine game outcome and probability, the use of one hot encoding for team and opponent identity altered these explanations. In use for both models, SHAP correctly identified win percentage for both teams as one of the most meaningful features for outcome predictions, while LIME primarily references team identity.

Discussion and Conclusion: While women's basketball is rapidly gaining popularity in the United States, it remains under-represented in similar research initiatives. This disparity influenced my decision to look at women's basketball specifically, as there are differences in the landscape of the sport compared to men's basketball.

This project highlighted the difficulty of capturing real-world nuance in a data set, and how models are highly dependent on specific project objectives and data. While the models both performed better than random (F1 score higher than 0.5), and logistic regression performed almost identically to an LLM, there is still room for improvement with data procurement as well as model complexity. Future efforts may include the addition of team statistics, or player specific statistics where each player is assigned to their correct team and given a binary value for injury status.

Additional figures and analysis are available in the project repository on GitHub.